

Supporting Information Appendix for

Direct evidence of surface exposed water ice in the lunar polar regions

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Null Hypothesis Tests

A null hypothesis test was performed to show that the three ice-like absorptions in M^3 data co-occur, indicating the presence of water ice. The null hypothesis was that the prediction of a third absorption by knowing the presence of the other two was insignificant. If the chi-squared value of our data was significantly greater than the critical chi-squared value at a given confidence level (i.e. 99%), then the null hypothesis was rejected. M^3 data were searched and subset into three categories: pixels showing absorptions near 1.5 and 2.0 μm , 1.3 and 1.5 μm , 1.3 and 2.0 μm , respectively. In each category there are two conditions that the remaining (third) ice absorption is either observed or not. Chi-squared values were calculated and compared with the chi-squared distribution at 99% confidence level, and the null hypothesis test results are presented in SI Appendix, Table S2 & S3. For each category we also calculated the average spectra of M^3 pixels identified using only two out of the three potential ice absorptions, and in each case the third diagnostic feature was shown to be present, further validating that all three absorptions occur together in a subset of M^3 pixels.

Diviner Temperature, LOLA Albedo, and LAMP Band Ratio of Ice Bearing Pixels

The final subset of potential ice-bearing pixels in the M^3 dataset are expected to occur at locations with temperature less than 110 K ², where ice is theoretically stable over long timescales, and these pixels are also expected to coincide with enhanced LOLA albedo values ³. Annual maximum bolometric

temperature maps derived from LRO Diviner data (~240 m/pixel) were spatially resampled ⁴ to M³ resolution (~280 m/pixel) to assess local surface temperature values. Similarly, M³ data were spatially resampled to the lower resolution LOLA albedo map ⁵ (~500 m/pixel) to examine relationships between these independent datasets. It has been suggested that micro cold traps (sub-pixel size) may exist in the lunar polar regions ⁶ that cannot be resolved by Diviner temperature data, and some of the ice exposures identified in the M³ data might exist in such locations. If ice is present in such micro cold traps, Diviner temperatures may overestimate actual values for ice-bearing surfaces observed in M³ pixels ⁶. Finally, the 'off' and 'on' band ratio map of LAMP data (250 m/pixel) that is used as an indicator of possible H₂O ⁶ was downgraded to the spatial resolution of M³ data at 280 m/pixel to compare these two datasets, both of which are sensitive to the uppermost fraction of the lunar surface. Median values for 9*9 pixel groups of LAMP data were used for this comparison, as suggested in ⁶.

Restrictions on the M³ Wavelength Range Used in this Study

In this study we examined the M³ data between 1.0 and 2.5 μm to search for potential water ice exposed at the lunar surface in cold traps. Below we explain the rationale of focusing on this wavelength range instead of using all M³ spectral bands (i.e., the exclusion of spectral bands between 0.46 – 1.0 μm , and 2.5 – 2.98 μm).

1. Lower Signal to Noise Ratios (SNR) and Apparent Lack of Strong Ice Absorptions at the Two Wavelength Ranges (0.46 – 1.0 μm and 2.5 – 2.98 μm)

The SNR of M^3 bands near the 1.5 μm water ice absorption are around 190 (red dashed line in Fig. S9), whereas most of the bands near the 3 μm wavelength region have SNR values near 150 (light blue dashed line in Fig. S9). Similarly, spectral bands at wavelengths shorter than 1 μm also show SNR values of ~ 150 . The highest SNR of the M^3 data at the polar region is near 250 (cyan dashed line in Fig. S9). It is noteworthy that the SNR value of the 1.5 μm region is 40 higher than bands in the 3 μm region (16% of the highest SNR value). The averaged M^3 spectra derived for both the north and south poles show a systematic increase in reflectance beyond band 76 (indicated by a pink dashed line Fig. S9), which is consistent with the decrease of SNR from band 77 to 85. This indicates the significant increase of noise level with decreasing of SNR and near the wavelength edge of the detector. As such, the 1.5 μm bands in M^3 data should be more reliable than the 3 μm bands (bands 77-85) although band 77 exhibits a slightly higher SNR value than that of the 1.5 μm bands. The reflectance values from band 77 to 85 were not presented in this paper because of the low SNR of these channels and their unreliability in measuring true surface reflectance properties.

The 3 μm absorption of pure water ice begins near 2.62 μm (band 76 for M^3 , pink dashed line in Fig. S9) and even spectra of pure ice exhibit only small variations from band 74 to band 76 (2.54 - 2.62-micron) (Fig. S9). Therefore, the reflectance values from band 74 to 76 were not shown in the main text because of their small variation in reflectance compared with that of band 73 (2.5 μm).

The SNR values from band 1 to band 22 (0.46 - 1.0 μm) of M³ data are around 50 lower than those of the 1.5 μm bands (Fig. S9). In addition, water ice spectra only exhibit two very weak overtone absorptions near 0.8 and 0.9 μm . When low abundance of ice (i.e., < 30%) is mixed with dry lunar regolith, these features are expected to be too weak to be detected (Fig. S10b). Because of such weak absorptions at these wavelengths, coupled with low SNR of M³ data in this spectral region, we do not focus on reflectance values from band 1 to band 22 of M³ data in this study.

2. Uncertainties in the Presence/Absence of a Water Ice Absorption Near 3 μm from Different Processing Approaches

To further examine whether or not the 3 μm feature is present we examined the original radiance data and compare spectra for non-ice pixels in shadowed regions with ice-bearing pixels in shadowed regions. Interestingly, the 3 μm absorption *does* appear in ratios of these radiance spectra (and the same would be true for I/F spectra because the processing from radiance to I/F is the same for all pixels). When the spectra of our detected ice-bearing pixels are ratioed to the spectra of ice-free pixels that are also in the cold traps we observe all of the H₂O features, including a decrease in reflectance towards 3 μm (Figure S11a).

The other spectral ratios shown in this work were of ice-bearing pixels in the shadowed regions relative to pixels in illuminated regions. We have determined that the reflectance spectra of illuminated regions originally used as a denominator in equation (1) contain a 3 μm absorption due OH (Figure S12), as is common at

high latitudes due to solar wind implantation e.g., ^{7,8,9}. In contrast, the spectra of ice-free pixels in the cold traps exhibit no such absorptions near 3 μm (Figure S12), likely because of low solar wind flux in these areas.

This is an interesting result, and the apparent lack of a 3 μm feature in the other ratio spectra is likely an artifact of dividing two spectra that both exhibit absorptions at these wavelengths, thus they are cancelled out in the ratio process (and because the OH feature in illuminated regions is so strong it may in fact explain the observed upturn near 3 μm in ratios such as those shown in Figure S9). The ratios of ice-bearing and ice-free pixels that are both from the cold trap regions do not suffer from this issue, and these ratios exhibit all of the characteristic and expected water features (Figure S11a).

Despite this result, both of the approaches of deriving the spectra of ice-bearing pixels are hindered by issues near the 3 μm band due to low SNR. This can be seen in Figure S12, where it is shown that reflectance spectra of illuminated regions near the poles exhibit a 3 μm absorption whereas reflectance spectra of ice-free pixels in the cold traps exhibit an upturn at long wavelengths due to low SNR. As such, the three ice-like absorptions near 1.3, 1.5, and 2.0 μm are more robust than the 3 μm absorption because they are present and remain unchanged regardless of which method is used in deriving the ratio spectra of ice-bearing pixels.

Another issue we considered is the weakening of ice absorptions when pure ice is mixed with 'dry' lunar regolith. As shown in the modeled mixture spectra in Figure S11b, the 3 μm band can be suppressed more strongly than the shorter

wavelength features in a nonlinear mixture. The spectra in Figure S11b were generated using a lab spectrum of pure water ice and a representative M³ spectrum for 'dry' lunar regolith. In summary, the apparent lack of a 3 μ m absorption is complicated by the offsetting effects of the presence of OH in illuminated regions and low SNR for shadowed regions. Thus, while the feature is present in certain ratio methods its presence is not as robust as the other absorption features.

Tables

Table S1. Characteristics of ice absorptions at the region of 1.0-2.5 μm adopted from Table 1-3 in ¹⁰.

Absorptions (μm)	Shoulders (μm)		Centers (μm)	
	Min	Max	Min	Max
1.1	0.900	1.100	1.028	1.048
1.3	1.130	1.350	1.242	1.323
1.5	1.420	1.740	1.503	1.659
2.0	1.820	2.200	1.945	2.056

Table S2. Results of a null hypothesis test for examining whether the prediction of a third ice-like absorptions by knowing the other two is significant in the southern polar region.

The southern polar region				
A Null hypothesis: The prediction of a third ice-like absorption by knowing the other two is insignificant.				
Observed frequency (<i>f_o</i>)	Known absorptions			
Able to predict the third absorption?	1.5 & 2.0 μm	1.3 & 1.5 μm	1.3 & 2.0 μm	Total
Yes	1356	1356	1356	4068
No	1860	3744	708	6312
Total	3216	5100	2064	10380
Expected frequency (<i>f_e</i>)	Known absorptions			
Able to predict the third absorption?	1.5 & 2.0 μm	1.3 & 1.5 μm	1.3 & 2.0 μm	
Yes	1260.4	1998.7	808.9	
No	1955.6	3101.3	1255.1	
Chi squared value = (<i>f_e</i> - <i>f_o</i>) ² / <i>f_e</i>	Known absorptions			
Able to predict the third absorption?	1.5 & 2.0 μm	1.3 & 1.5 μm	1.3 & 2.0 μm	
Yes	7.26	206.68	370.04	
No	4.68	133.20	238.48	
Chi squared value (Sum of cell values in the chi squared table)	960.34			
Degree of freedom (DoF)	2			
Critical chi squared value	9.21 at 99% confidence level & DoF = 2			
960.34 >> 9.21, The null hypothesis can be rejected at 99% confidence level.				

Table S3. Results of a null hypothesis test for examining whether the prediction of a third ice-like absorptions by knowing the other two is significant in the northern polar region.

The northern polar region				
A Null hypothesis: The prediction of a third ice-like absorption by knowing the other two is insignificant.				
Observed frequency (<i>f_o</i>)	Known absorptions			
Able to predict the third absorption?	1.5 & 2.0 μm	1.3 & 1.5 μm	1.3 & 2.0 μm	Total
Yes	408	408	408	1224
No	758	1458	343	2559
Total	1166	1866	751	3783
Expected frequency (<i>f_e</i>)	Known absorptions			
Able to predict the third absorption?	1.5 & 2.0 μm	1.3 & 1.5 μm	1.3 & 2.0 μm	
Yes	377.3	603.7	243.0	
No	788.7	1262.3	508.0	
Chi squared value = (<i>f_e</i> - <i>f_o</i>) ² / <i>f_e</i>	Known absorptions			
Able to predict the third absorption?	1.5 & 2.0 μm	1.3 & 1.5 μm	1.3 & 2.0 μm	
Yes	2.50	63.47	112.06	
No	1.20	30.36	53.60	
Chi squared value (Sum of cell values in the chi square table)	263.18			
Degree of freedom (DoF)	2			
Critical chi squared value	9.21 at 99% confidence level & DoF = 2			
263.18 >> 9.21, The null hypothesis can be rejected at 99% confidence level.				

Figures

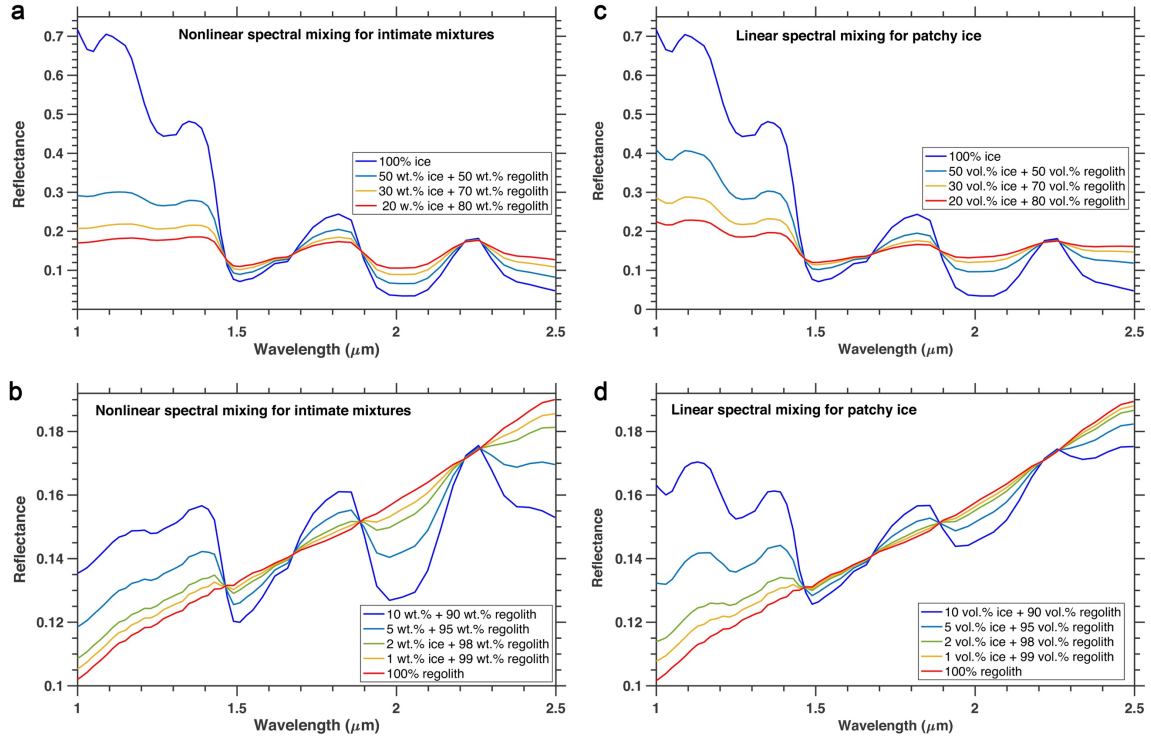


Fig. S1. Modeled reflectance spectra of ice and lunar regolith mixtures using Hapke's radiative transfer model (a, b) ¹¹ and a linear mixing method (c, d), the spectrum of pure water ice is from the USGS spectral library, the spectrum of lunar regolith is the average M³ spectrum in the lunar polar regions (75° N/S – 90° N/S), densities of ice and lunar regolith used in the Hapke modeling are 0.92 g/cm³ and 1.80 g/cm³, respectively, particle sizes of ice and regolith are both set as 20 μm .

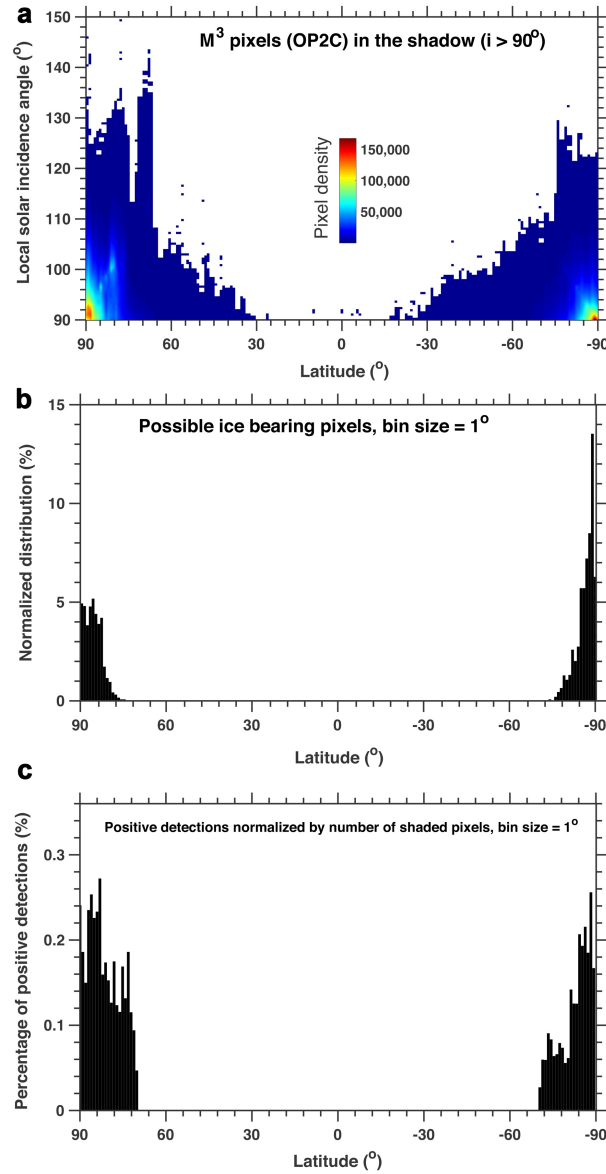


Fig. S2. Detection results of ice bearing pixels using the global M³ data; a. the distribution of shaded pixels with latitudes; b. the distribution of ice bearing pixels, that only show at latitudes $> \sim 70^\circ$ N/S, $> 90\%$ ice bearing pixels locate at latitudes $> \sim 80^\circ$ N/S; c. the percentage of shaded pixels shows ice absorption features at each latitude, which is similar in the south and north polar regions.

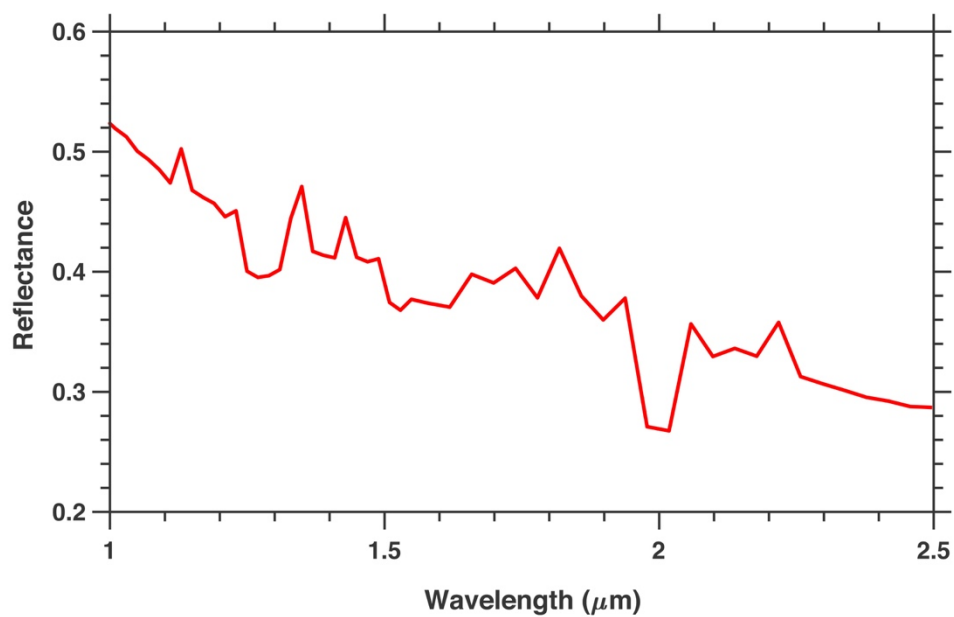


Fig. S3. The spectral ratios between the average spectrum of random data (~65 million data points) passing our criteria for detecting ice and the average spectra of M³ data at illuminated regions near both poles (75° N/S – 90° N/S).

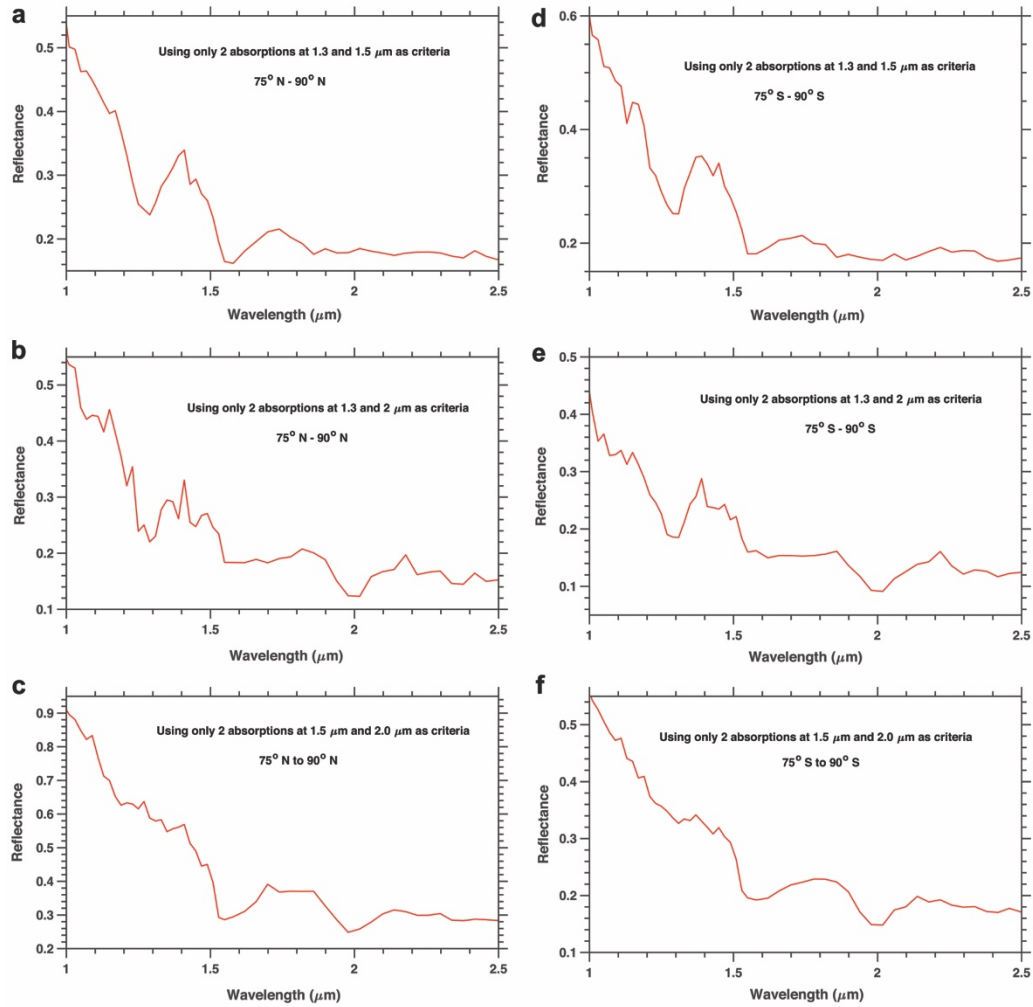


Fig. S4. The average spectra of M^3 pixels passing the criteria of any two of the three ice absorptions near 1.3 μm , 1.5 μm , and 2.0 μm , the spectra were divided by the average spectra of illuminated regions in both poles to eliminate the spectral information inherent from the light source; a-c. average spectra of M^3 pixels passing the criteria of showing absorptions at 1.3 μm and 2.0 μm , 1.5 μm , and 2.0 μm , 1.3 μm and 1.5 μm , respectively in the northern polar region (75° N – 90° N); d-f. average spectra of M^3 pixels passing the criteria of showing absorptions at 1.3 μm and 2.0 μm , 1.5 μm , and 2.0 μm , 1.3 μm and 1.5 μm , respectively in the southern polar region (75° S – 90° S).

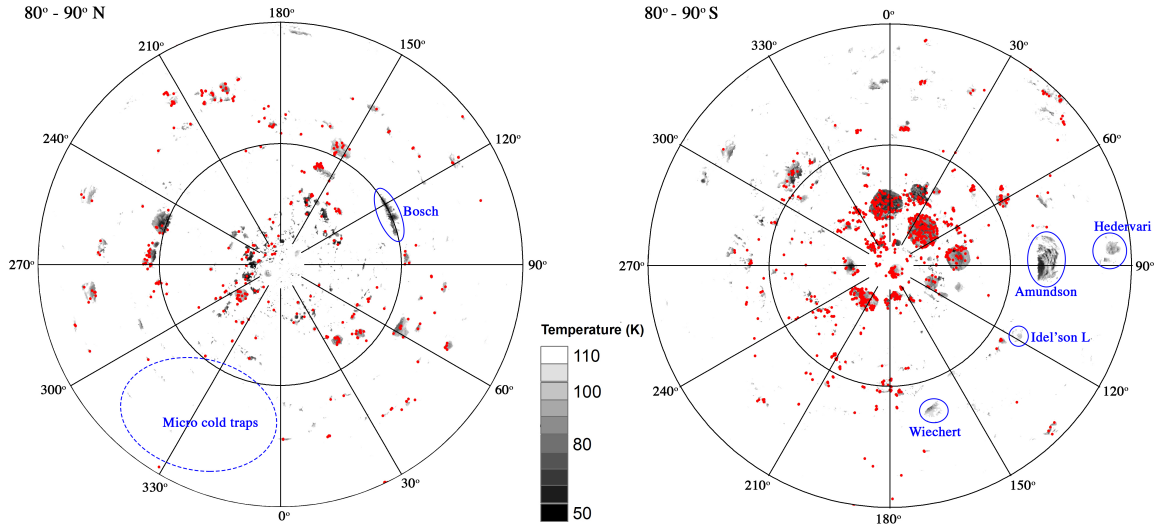


Fig. S5. Ice exposures overlain on the Diviner annual maximum bolometric temperature, only temperatures no greater than 110 K are colored from black to white, cold traps not showing ice exposures at crater Bosch and some micro cold traps along the 330° longitude are circled in blue on the northern pole, cold traps not showing ice exposures at craters Hedervari, Amundson, Idel'son L, and Wiechert are circled in blue on the southern pole.

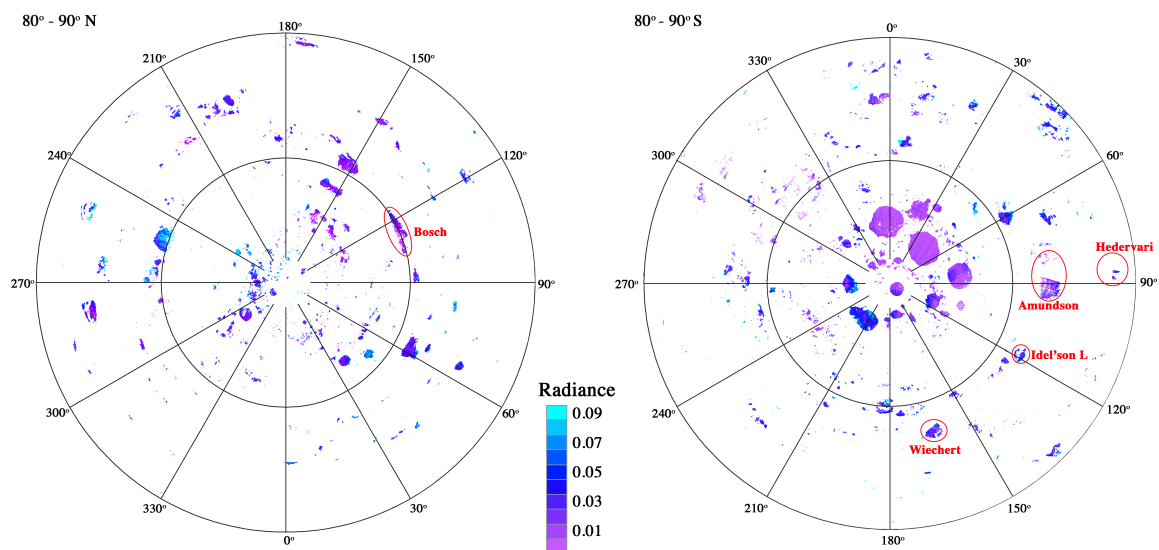


Fig. S6. The mean radiance of M³ data between 1.0 and 2.5 μm at all cold traps (< 100 K) near the north (left) and south (right) pole, respectively.

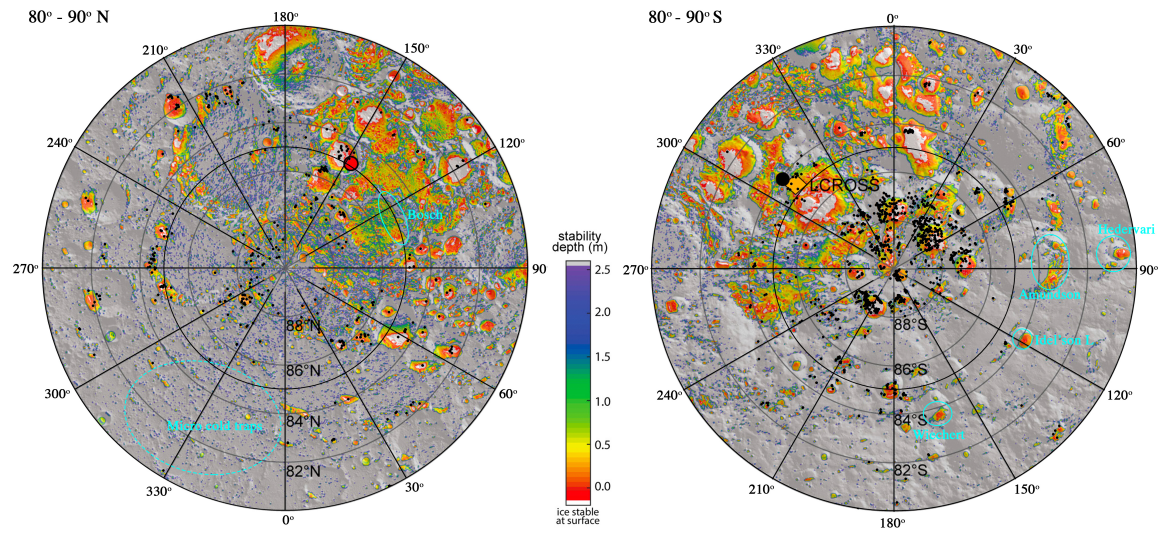


Fig. S7. Ice exposures (black dots) and cold traps not showing ice (cyan circles) are overlain on the map of modeled ice stability depth when the Moon is hypothesized on its paleo-axis ²².

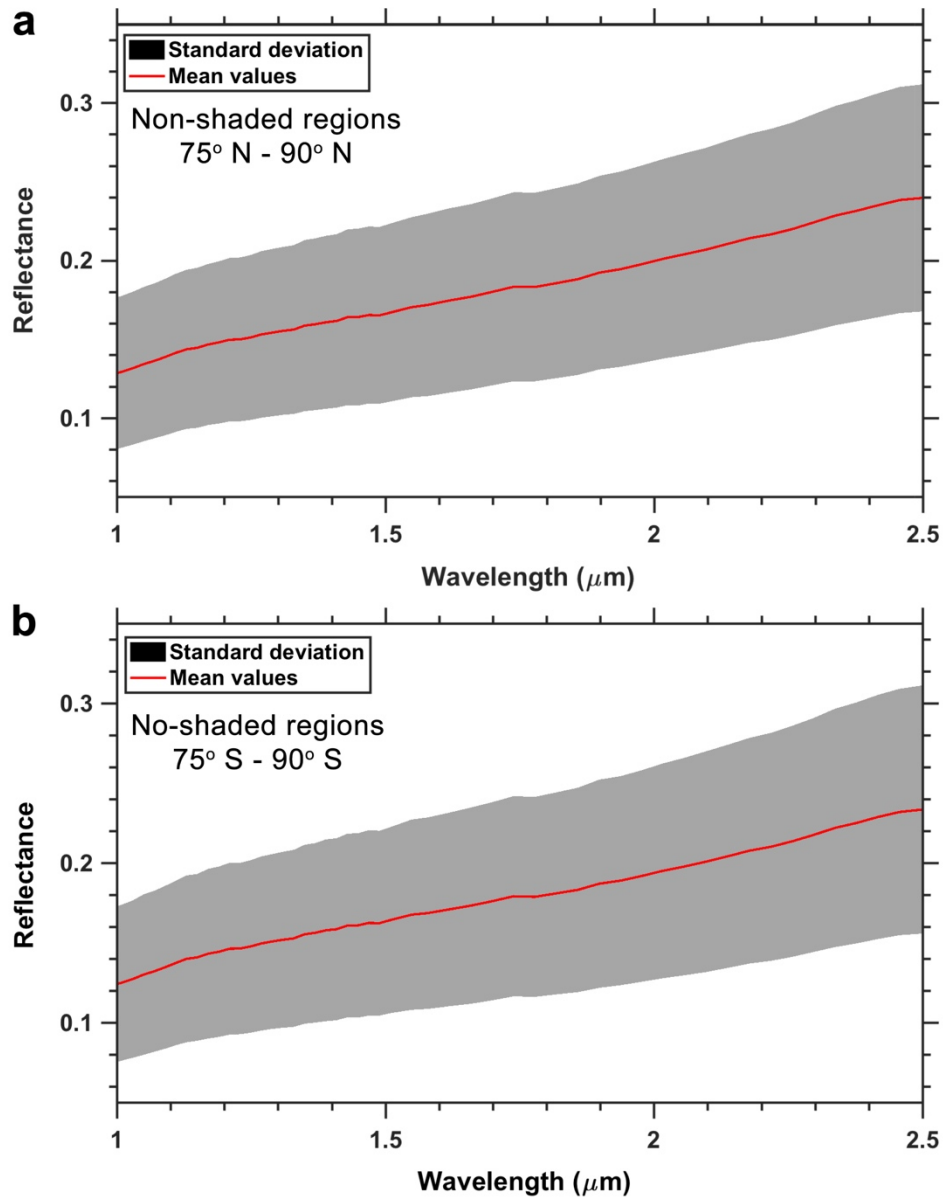


Fig. S8. The average spectra and standard deviations of M^3 data at illuminated regions in the northern (a) and southern (b) polar regions, only the spectral bands between 1.0 μm and 2.5 μm were used in this study.

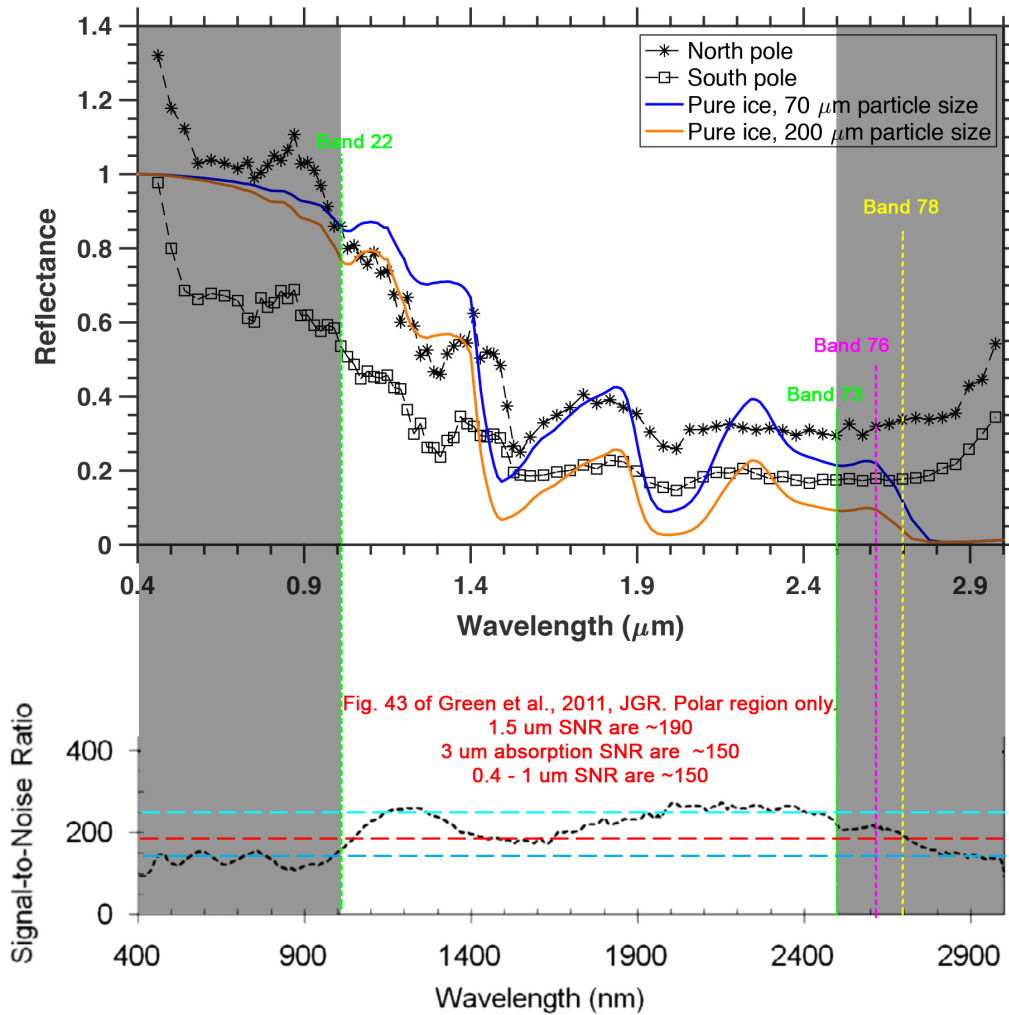


Fig. S9. The full wavelength range M^3 spectra of ice-bearing pixels compared with pure ice spectra and the overall signal to noise ratio (SNR) of the M^3 data at the polar regions ¹.

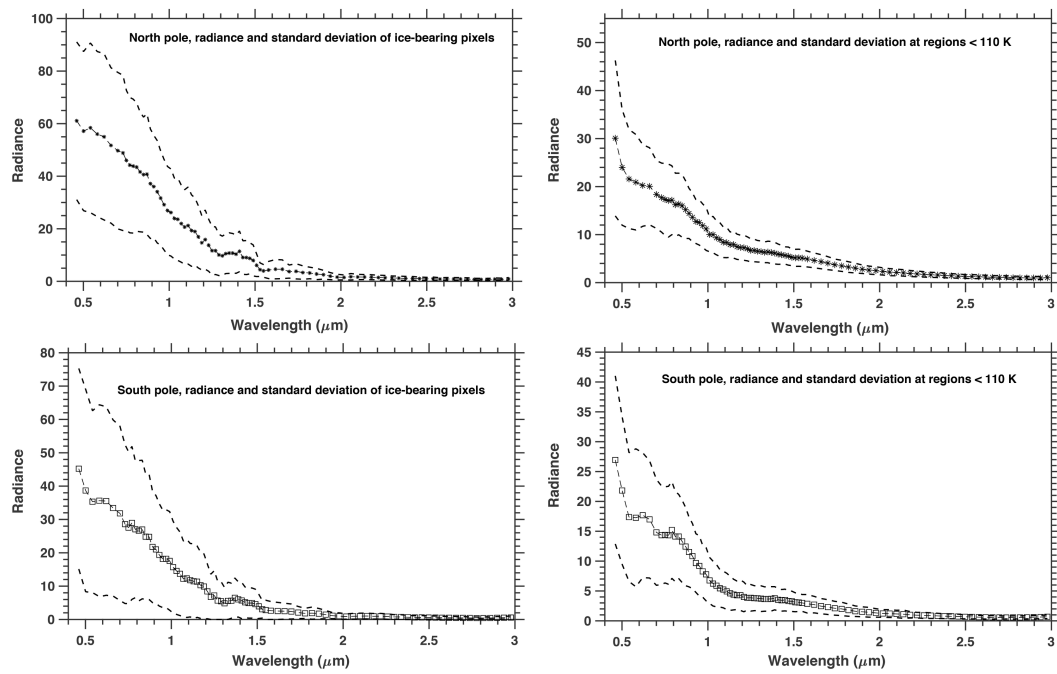


Fig. S10. M³ radiance and standard deviation values for ice-bearing and ice-free pixels at cold traps near the north and south poles.

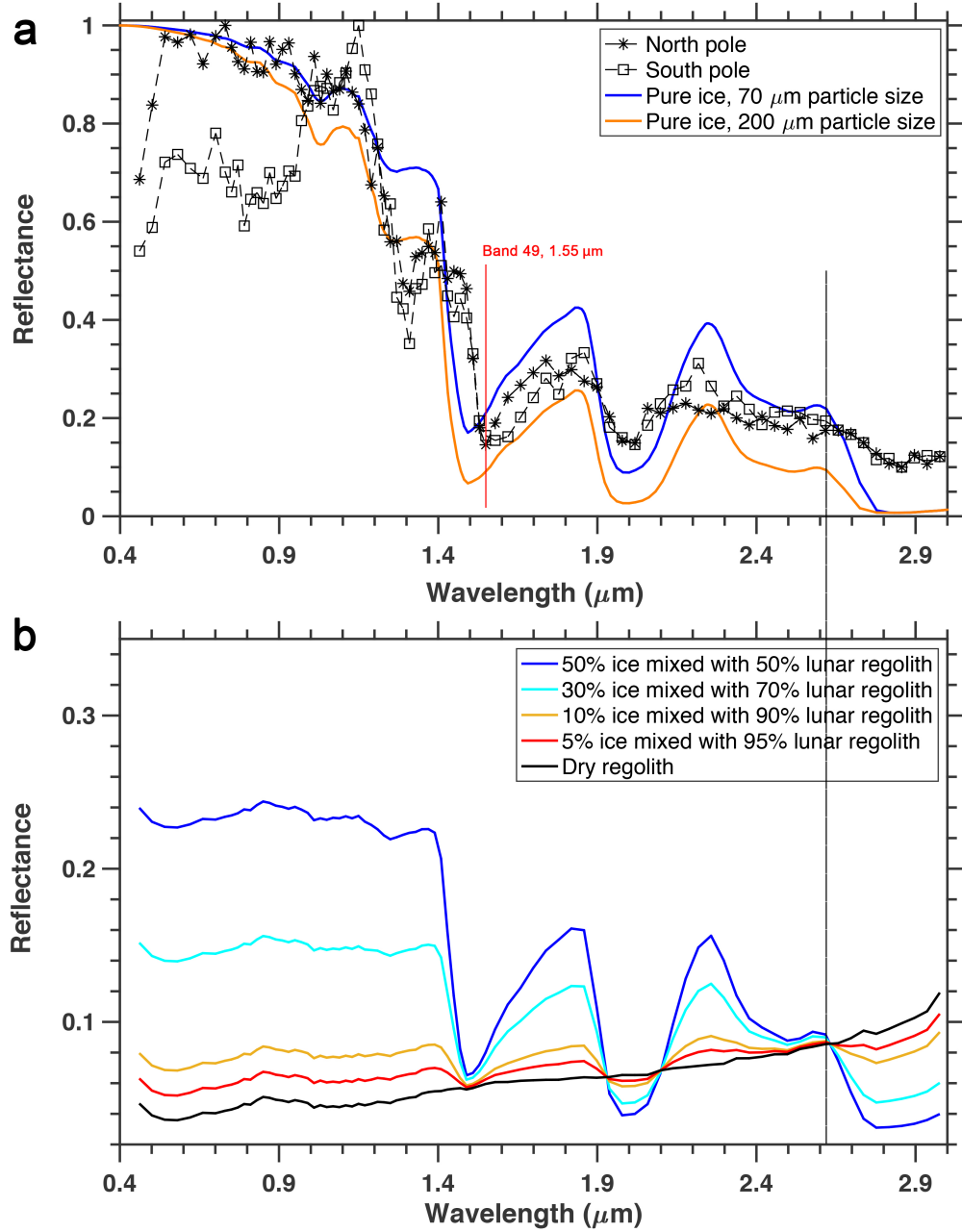


Fig. S11. a. Ratio spectra of ice-bearing pixels and ice-free pixels compared with the spectra of pure ice; b. nonlinear mixing of pure ice with lunar regolith; the spectra of lunar regolith are shown in Figure S12.

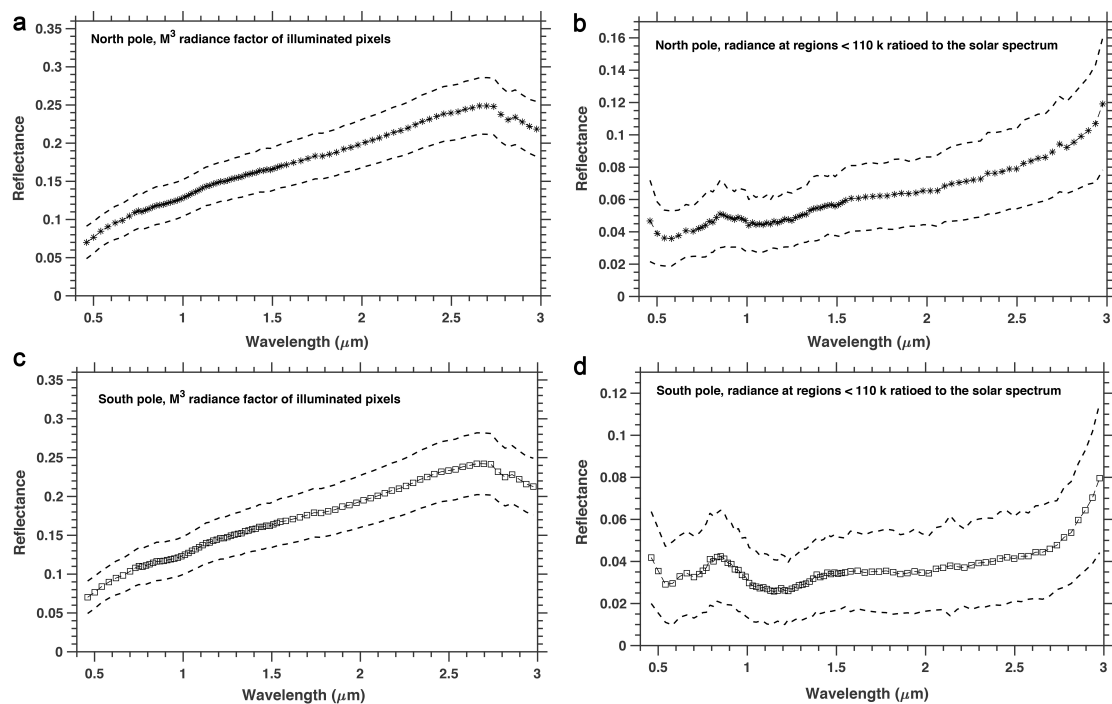


Fig. S12. The reflectance spectra of illuminated pixels near the north (a) and south (c) pole, respectively; the reflectance spectra of ice-free pixels in the cold traps near the north (b) and south (d) pole, respectively. The dashed lines are the standard deviation.

References:

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